

Quantized Conductance Report

Katie Carlson

Contents

1	Introduction and Theory	2
2	Equipment and Procedure	3
3	Data and Calculations	4
4	Error Analysis	11
5	Conclusion	12
6	Appendix	13

1 Introduction and Theory

Just as photons are quanta of light, this experiment aims to find the quanta of conductance. Conductance, denoted by the letter G , is the inverse of resistance, given by $G = 1/R$.

To derive G_0 , the quanta of conductance, we begin by considering a wire carrying a current $I = veN/L$, where v is the velocity of the moving electrons, e is the standard electron charge, N is the number of electrons per unit length of the wire, and L is the length of the wire.

Furthermore, we know that eV is the drop in the potential energy ΔE of an electron that goes from one end of the wire to the other.

We can then rearrange the formula $V = IR$ to obtain:

$$G = \frac{ve^2N}{L\Delta E} \quad (1)$$

To evaluate this expression, we consider the allowed quantum states of the electrons in a one-dimensional conductor. For electrons confined to a length L , the allowed wave numbers are quantized according to

$$k_n = \frac{n\pi}{L},$$

where n is an integer. Each allowed k -state occupies a length π/L in k -space, so the number of available states per unit length in k -space is L/π .

Using the free-electron dispersion relation

$$E = \frac{\hbar^2 k^2}{2m},$$

we see that near the Fermi energy only electrons within an energy window $\Delta E = eV$ contribute to conduction. The number of states in this energy window is therefore determined by the density of states at the Fermi level.

The number of electrons per unit length contributing to transport is

$$N = \frac{2}{\pi} \frac{\Delta k}{L},$$

where the factor of 2 accounts for spin degeneracy. Using the relation between energy and wave number,

$$\Delta E = \hbar v_F \Delta k,$$

where v_F is the Fermi velocity, we can substitute $\Delta k = \Delta E/(\hbar v_F)$ into the expression for N . Substituting this result into Equation (1) gives

$$G = \frac{ve^2}{L\Delta E} \left(\frac{2L}{\pi} \frac{\Delta E}{\hbar v_F} \right).$$

Since the electrons contributing to conduction travel at approximately the Fermi velocity, we set $v = v_F$. After cancelling common terms, we obtain

$$G = \frac{2e^2}{\pi\hbar}.$$

Recalling that $\hbar = h/2\pi$, this becomes

$$G = \frac{2e^2}{h}.$$

This quantity,

$$G_0 = \frac{2e^2}{h},$$

is known as the quantum of conductance. Numerically,

$$G_0 = 7.75 \times 10^{-5} \text{ S}.$$

Therefore, in a one-dimensional conductor with perfect transmission, the conductance does not vary continuously, but instead occurs in discrete integer multiples of G_0 :

$$G = nG_0,$$

where n is the number of available quantum transport channels. This leads to a step-like behavior in conductance that this experiment seeks to observe.

2 Equipment and Procedure

The following equipment was used to perform the quantized conductance experiment:

- Haasoscope data acquisition system (Fast ADC channel enabled)
- Transimpedance amplifier circuit with feedback resistance $R_{\text{in}} = 100 \text{ k}\Omega$

- Gold wire break-junction apparatus
- Piezoelectric actuator for controlled mechanical displacement
- Arduino-based voltage source for applying bias voltage
- Computer with Python (NumPy, Pandas, Matplotlib) for data analysis
- Oscilloscope interface software for waveform visualization and data export

Experimental Setup and Procedure

Two gold wires were mounted in a mechanically controlled break-junction configuration. One wire was fixed while the other was attached to a piezoelectric actuator. A DC bias voltage V_{in} was applied across the junction using an Arduino-based voltage source.

The current flowing through the junction was converted into a voltage signal using a transimpedance amplifier with feedback resistance $R_{in} = 100 \text{ k}\Omega$. The output voltage V_{out} was recorded using the Haasoscope Fast ADC channel in rolling acquisition mode.

For each applied bias voltage (ranging from 100 mV to 1 V, increasing by increments of 100 mV), the following steps were performed:

1. The gold wires were brought into contact and then slowly separated using the piezo actuator.
2. During separation, the Haasoscope recorded the output voltage trace $V_{out}(t)$.
3. Two minutes worth of frames were recorded and saved as csv files for analysis.

3 Data and Calculations

To calculate our experimental value of G_0 , we'll first use the ratio of $V_{in}/V_{out} = -R_2/R_{in}$ to calculate R_2 , the resistance of the gold wire. We can then use the formula $G = 1/R_2$ to obtain G . We do not expect our experimental value of G_0 to change based on the input voltage because there is no explicit dependence on V_{in} in the relationship.

We examine "frames" of data that look at output voltage over a minuscule period of time, where the piezo crystal is driving the wires out of contact with each other. In each of the ten files, we selected a frame where "steps" were visible, a clear indication of the quantization of conductance. We then calculated G and the ratio G/G_0 , and created a histogram of all the values of G/G_0 and how many times they occur (the counts).

To denoise our data, we utilized the wavelet transform technique in Python. First, we performed a Discrete Wavelet Transform on the signal using the Daubechies-4 wavelet. Then, we calculated a threshold value from the standard deviation of the detail coefficients and removed coefficients smaller than the threshold, effectively reducing small fluctuations from noise. Lastly, we reconstructed the signal using an Inverse Wavelet Transform to ultimately produce a smoothened version of the signal for analysis.

In the following graphs, we plot time steps (20 microseconds) against the output voltage (volts) for the denoised data.

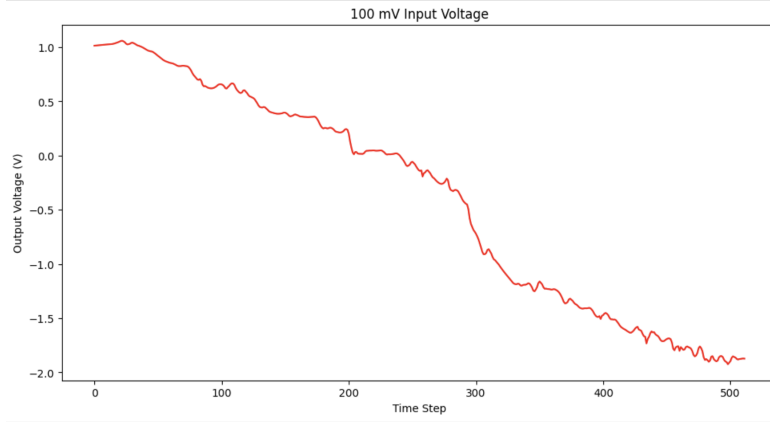


Figure 1: 100 mV Input Voltage

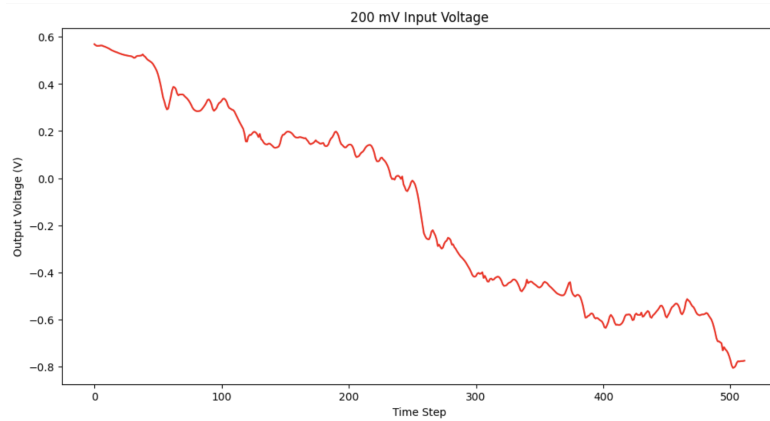


Figure 2: 200 mV Input Voltage

1

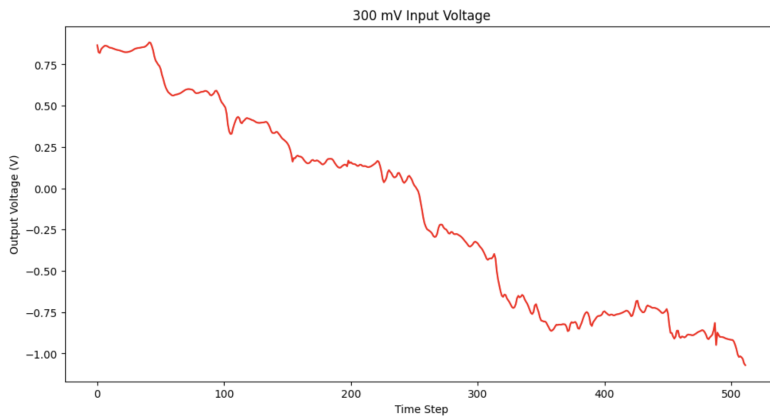


Figure 3: 300 mV Input Voltage

¹We referenced one of the papers cited in the lab manual, E. L. Foley; D. Candela; K. M. Martini; M. T. Tuominen, An undergraduate laboratory experiment on quantized conductance in nanocontacts <https://pubs.aip.org/aapt/ajp/article/67/5/389/1045772/Anundergraduate-laboratory-experiment-on>. We will include a figure from that paper in our appendix.

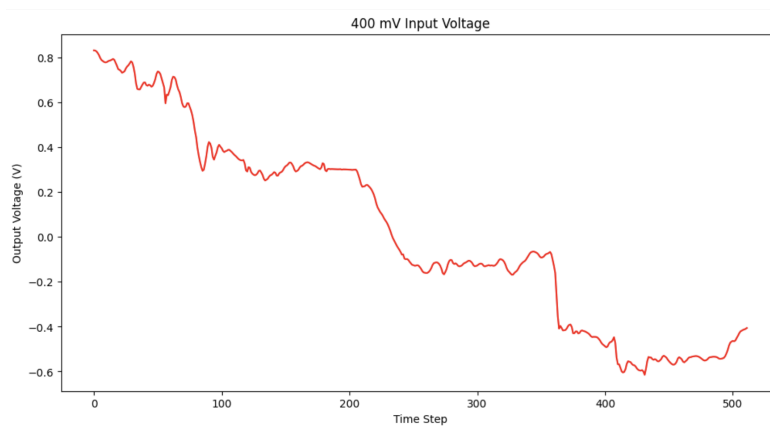


Figure 4: 400 mV Input Voltage

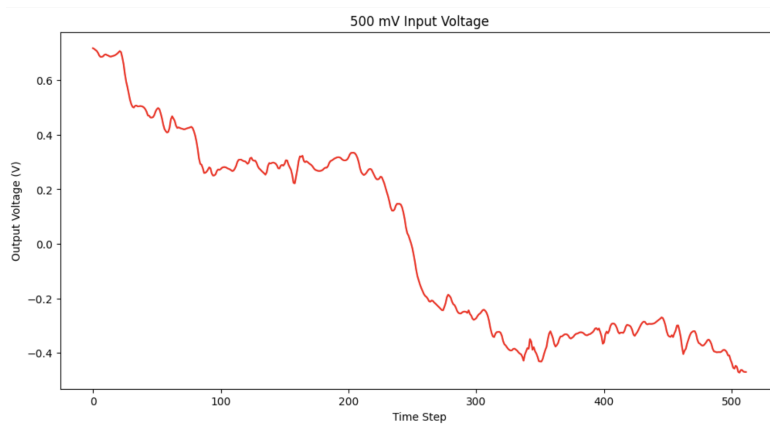


Figure 5: 500 mV Input Voltage

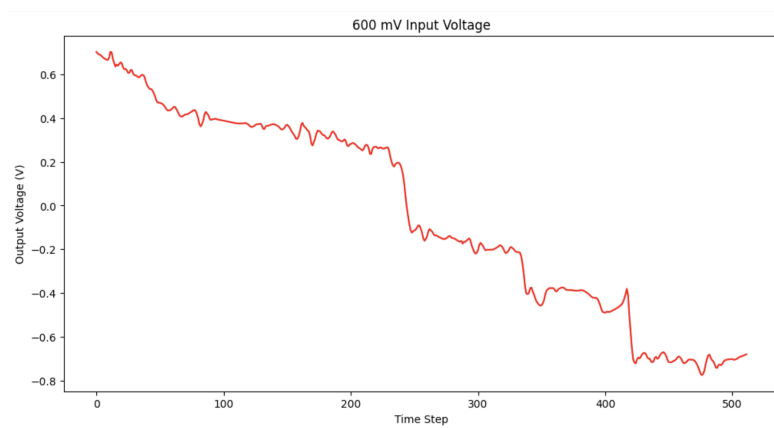


Figure 6: 600 mV Input Voltage

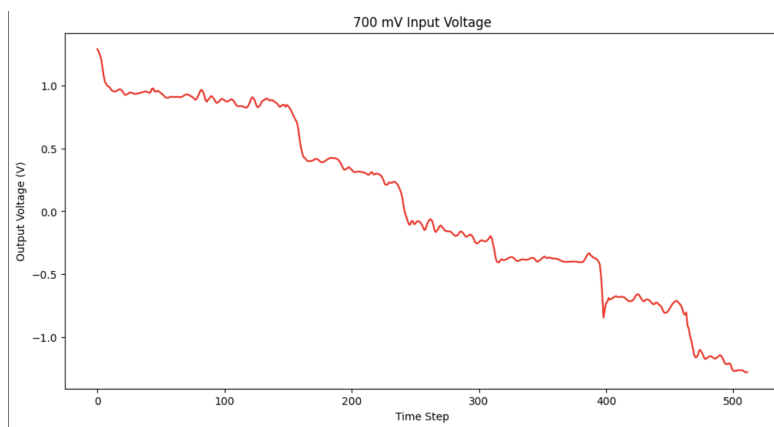


Figure 7: 700 mV Input Voltage

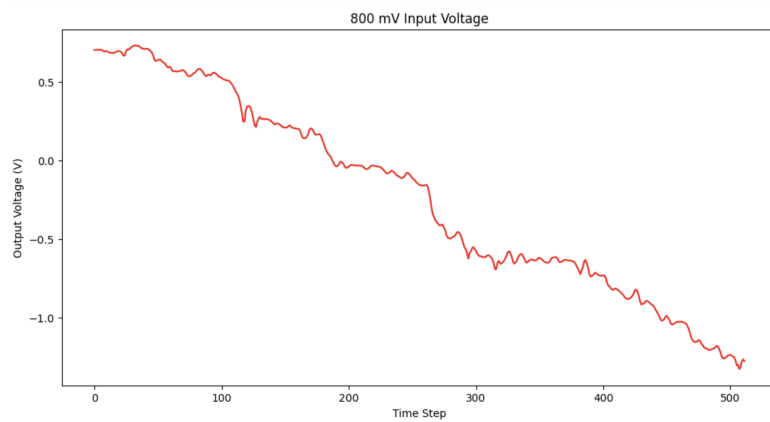


Figure 8: 800 mV Input Voltage

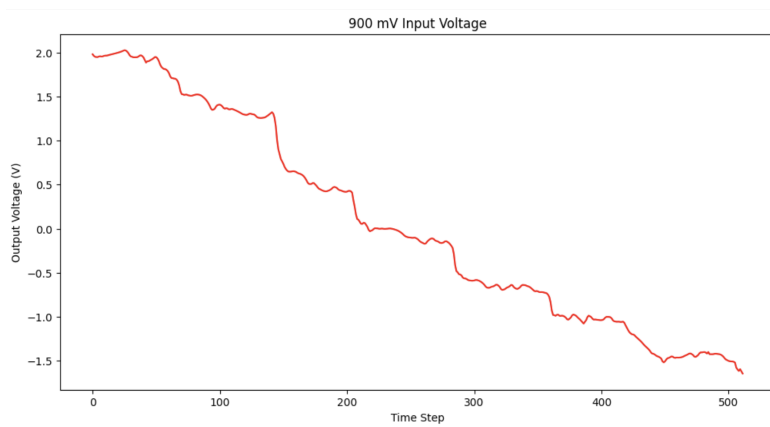


Figure 9: 900 mV Input Voltage

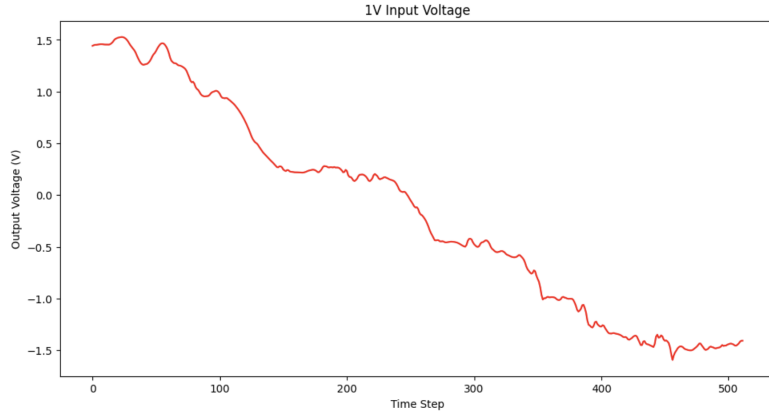


Figure 10: 1 V Input Voltage

We use the formula for G we derived in the theory section to calculate the conductance. Then, aggregating the data from all ten traces together, we created a histogram plotting the ratio G/G_0 against the counts. In the histogram chart, we expect to see maxima at integer values, as we expect conductance to manifest in integer multiples of the base quanta of conductance G_0 . We obtained the following graph: ²

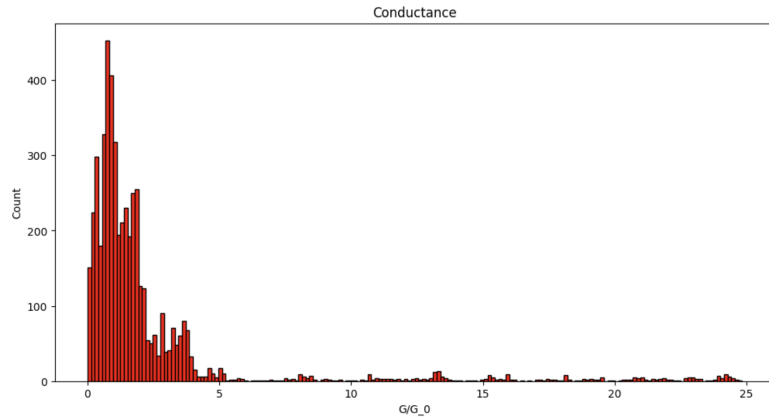


Figure 11: Histogram of conductance ($2e^2/h$) v.s. count

The maximum value occurs at 0.69, and we can see several other local

²The code we used for data analysis will be included as a separate attachment to this report.

maxima. We expand on why the maximum isn't 1 like we expected in the error analysis section.

4 Error Analysis

We will use the error propagation formula to gauge how much room for error there was in our experiment:

$$\sigma_F = \sqrt{\left(\frac{\partial F}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial F}{\partial y}\right)^2 \sigma_y^2}$$

Given

$$G = \frac{V_{out}}{-R_1 V_{in}}$$

We have

$$\sigma_G = \sqrt{\left(\frac{\partial F}{\partial V_{out}}\right)^2 \sigma_V^2}$$

Here, we assume the only variable with error is V_{out} , due to noise in the data. We assume that both the values of R_1 and V_{in} are, for our purposes, very accurate due to the precision of the NYU lab equipment.

With an error σ_V of 10mV, we obtain

$$\begin{aligned} \sigma_G &= \sqrt{\left(\frac{-1}{R_1 V_{in}}\right)^2 \sigma_V^2} \\ &= 2 * 10^{-8} \text{S} \end{aligned}$$

With the textbook G_0 set at $7.75 * 10^{-5}$ S, this gives us an error margin of 0.25%.

However, given that we were off by 31% even with noise suppression, we must consider a large source of systematic error in our data. We hypothesize that there is much more resistance in the equipment than what we accounted for, as the value of R_1 given in the manual is only for the resistance of the operational amplifier. With more resistance in the rest of the equipment, it makes sense for conductance (which is inversely related to resistance) to be

smaller, explaining why our experimental G is noticeably smaller than the textbook G_0 .

5 Conclusion

While we do have noticeable error margins, we can still clearly see the quantization of conductance at the moment the wires break contact with each other. Repeated ten times for varying values of V_{in} , it's apparent that the value of conductance tends to cluster around integer multiples of G_0 , with G_0 itself being the most populated.

6 Appendix

We referenced Figure 2 from the paper by Foley et al. for our data and calculation section:

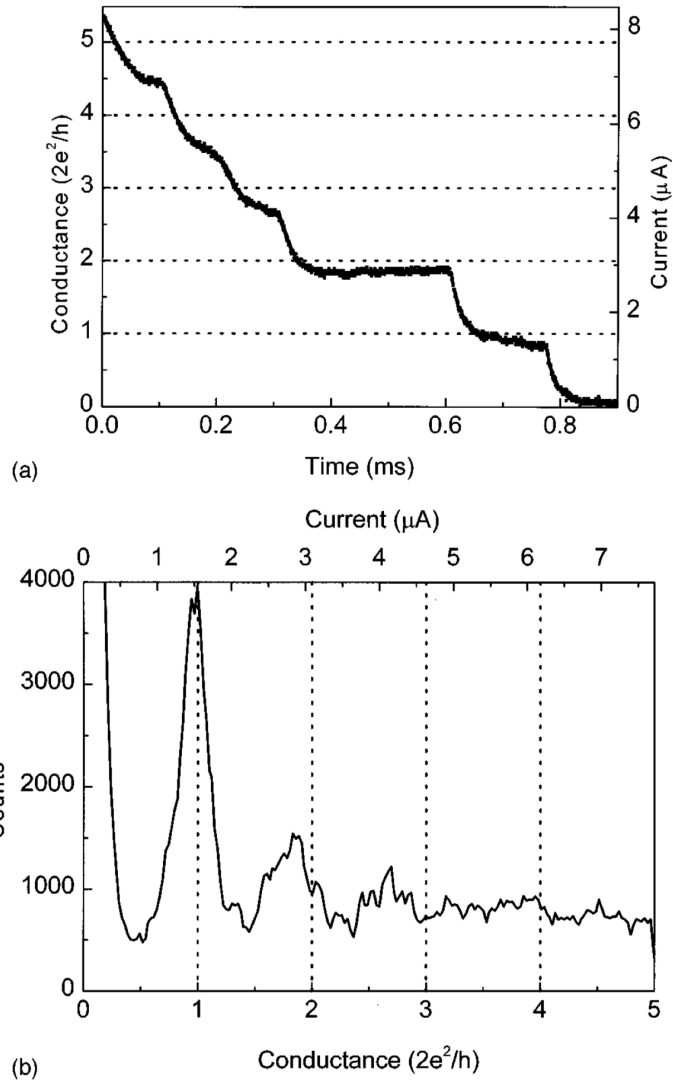


Figure 12: The upper graph shows a trace while the lower graph shows the aggregated histogram of 104 traces